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Prevalence, subtypes, and comorbidity of DSM-5 insomnia disorder among adults in Beijing, China: a large-scale cross-sectional study

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Abstract

Background Despite the crucial distinction between insomnia symptoms and a diagnosed disorder, population-level studies based on contemporary criteria and clinical interviews are scarce. This study therefore examined the insomnia spectrum by assessing the prevalence, identifying subtypes, and exploring associations with sociodemographic factors and comorbid mental disorders for both conditions.

Methods This large-scale, community-based cross-sectional study was conducted in Beijing from October to December 2021. A sample of 10,778 adults was recruited via multistage stratified random sampling. Trained psychiatrists conducted standardized diagnostic interviews based on the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) to collect data on insomnia disorder and mental disorders. Descriptive analysis and weighted Rao-Scott chi-square tests, were performed using the Statistical Analysis System (SAS, version 9.4).

Results The weighted prevalence rates were 17.7% for subjective sleep problems, 10.7% for clinically assessed insomnia symptoms, and 3.0% for DSM-5–diagnosed insomnia disorder. Among insomnia subtypes, initial insomnia was most prevalent, both as a symptom (8.1%) and a disorder (2.6%). Furthermore, insomnia disorder prevalence varied by sociodemographics, being higher in females, older adults (≥ 60 years), and those with lower education. Among individuals with insomnia disorder, 31.3% had comorbid other mental disorders, particularly alcohol-related disorders (13.4%). Conversely, insomnia disorder was observed in 14.4% of individuals with other mental disorders, with the highest prevalence in depressive (21.4%) and anxiety disorders (19.2%).

Conclusions The marked disparity between prevalent insomnia symptoms and formal diagnoses, compounded by significant psychiatric comorbidity, mandates a public health shift toward population-level screening and early intervention. This imperative includes adopting standardized diagnostics and implementing integrated, transdiagnostic treatment models as a pivotal preventive strategy.

Keywords Insomnia disorder, Insomnia symptoms, Insomnia subtype, Comorbidity, DSM-5

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Background

Over the past decade, poor sleep health has emerged as a significant public health issue [1], with a meta-analysis of data from 38 countries revealing that the global prevalence of insomnia ranks among the highest [2]. In this context, it is crucial to recognize that in sleep medicine and epidemiological research, a fundamental nosological distinction separates the presence of insomnia symptoms from a formal diagnosis of insomnia disorder. Insomnia itself constitutes a symptom (self-reported sleep difficulty) or a sign (polysomnographic disturbance) [3]. In contrast, insomnia disorder is a medical condition defined by specific criteria: symptoms occur at a stipulated frequency, cause significant social or functional impairment, and are not better explained by other factors [4]. This differentiation is critical, as it guides accurate clinical assessment, ensures epidemiological precision, and directs appropriate intervention strategies.

Insomnia disorder progresses from initial sleep complaints, with its persistent symptoms driving a cascade of adverse effects—from cognitive and mood impairments to increased risks of comorbid conditions [5–7]. Furthermore, it generates substantial socioeconomic costs due to reduced work efficiency [8]. A comprehensive understanding of the prevalence of both insomnia symptoms and insomnia disorder is, therefore, fundamental for shaping effective interventions and allocating healthcare resources.

Large-scale epidemiological studies specifically diagnosing insomnia disorder remain scarce in China. However, numerous cross-sectional studies have estimated the prevalence of insomnia symptoms using standardized instruments. In terms of insomnia symptoms, studies from several Chinese regions have reported prevalence rates of 15%–49% in the general population, using sleep-related screening or severity scales [9–11]. However, a prevalence of 9.2% for clinically diagnosed insomnia disorder was reported in an epidemiological survey two decades ago in one Chinese province [12], which was based on the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) [13] and confirmed by psychiatrists. Representing a significant shift in nosology, the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) [14] reconceptualized insomnia as a distinct disorder rather than merely a symptom of other conditions [4]. Moreover, the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision (DSM-5-TR) [15] retains the same diagnostic criteria for insomnia disorder. However, epidemiological studies on insomnia disorder that are based on DSM-5 diagnostic criteria and confirmed through psychiatric interviews remain scarce, a situation that is particularly pronounced in China.

China's focus on insomnia is underscored by a series of public health policies. The “Healthy China 2030” [16] planning outline designated curbing the rise in insomnia prevalence as a key target, while the later “Pediatric and Mental Health Services Year” (2025–2027) [17] explicitly calls for the widespread creation of sleep clinics in medical institutions. There is, therefore, a pressing need to obtain high-quality, regionally representative data on the prevalence of insomnia symptoms and insomnia disorder in China. In an ideal epidemiological study, insomnia disorder should be defined by formal diagnostic criteria, assessed by trained professionals using structured interviews [18]. In response to this background, the Beijing Mental Health Survey (BMHS) was launched in 2021 by the Beijing Municipal Health Commission to bridge the existing gaps in Beijing's mental health research. This study aimed to investigate the prevalence of insomnia symptoms and disorder and to explore their associations with sociodemographic factors and comorbid mental health conditions. The findings provide critical baseline data that can help standardize epidemiological research on insomnia and inform more precise public health interventions.

Methods

The BMHS was a cross-sectional epidemiological survey of mental disorders that utilized a one-step procedure, in which trained psychiatrists directly employed the adapted Chinese version of the Structured Clinical Interview for DSM-5 Disorders, Research Version (SCID-5-RV) [19] to ensure diagnostic consistency and cross-cultural validity.

Study population

The target population for the BMHS comprised all community-dwelling Chinese adults living in Beijing. The sample size was estimated accounting for the complex sampling design of the BMHS:

$$N = \frac{Z_{1-\alpha/2}^2 \times p \times (1 - p)}{d^2} \times deff$$

The sample size calculation was based on key parameters derived from a previous study that applied the DSM-5 diagnostic criteria for insomnia disorder [20]: a prevalence (P) of insomnia disorder of 3.0%, a design effect ($deff$) of 2.0, a 95% two-sided confidence level, and a permissible error (d) of 0.15 P , yielding an initial estimate of nearly 10,500 individuals. After adjusting for a projected 70% effective response rate, the final sample size was set at 15,000.

From October to December 2021, we employed a four-stage, non-proportional stratified sampling method. To enhance sampling efficiency, a stratified design was

adopted. The sampling frame was first stratified into 16 layers according to Beijing's 16 administrative districts. The sample size for each stratum was determined based on its district-wide population, the populations of its constituent streets and towns, and its urban–rural distribution, all of which were sourced from Beijing's Seventh National Population Census.

In the first stage, streets or towns were selected from the districts using Probability Proportionate to Size (PPS) sampling. This process resulted in the selection of 2 to 7 streets or towns per district, culminating in a total of 50 streets and 20 towns being included in the survey.

The second stage involved selecting communities or villages from the 70 sampled streets or towns. Using the PPS method, three communities or villages were selected from each street or town, resulting in a total of 174 communities and 36 villages.

In the third stage, households were selected from the sampled communities or villages. Due to budget constraints, the target for completed interviews was set at 50 effective households per unit. Accordingly, 70 to 80 households were initially selected in each community or village, based on the estimated response rate derived from our previous survey experience.

The fourth stage entailed the selection of respondents from the sampled households using the Kish grid method [21]. As the survey was conducted via computer-assisted personal interviewing (CAPI), the eight Kish grid tables were embedded within the CAPI system. Respondents were then automatically selected by the computer. A total of 12,379 respondents were selected through this process. Those who met the following criteria were included: 1) Chinese nationality; 2) age ≥ 18 years; and 3) a minimum of six months' residence in Beijing during the previous year. Those who had communication-impeding hearing or visual impairments, or were pregnant (for ethical reasons), were excluded.

Procedure

In this large-scale epidemiological study, face-to-face interviews were employed as the primary data collection method. Diagnostic assessments were conducted through pen-and-paper personal interviews (PAPI), supplemented by CAPI or specific questionnaires. The integrated CAPI and PAPI modules within the electronic data-entry platform enabled real-time data uploading and timely quality checks by the quality control team.

A total of 113 psychiatrists were trained as interviewers in a nine-day standardized program to ensure inter-rater reliability. The training curriculum covered standardized procedures, administration of the SCID-5-RV, and effective interviewing techniques. Following mock interviews and a competency assessment, all psychiatrists were certified for fieldwork.

To evaluate feasibility and identify potential procedural challenges, a pilot study was conducted in two communities—one urban and one rural. Strategies to maximize response rates in the main survey included disseminating project information through community and social media channels, offering flexible interview scheduling (including evenings and weekends), conducting interviews in neighborhood offices to ensure participant comfort, and providing time-based compensation.

The survey procedure began with trained coordinators uploading pre-sampled household data to the electronic platform. Interviewers then visited selected households and, with assistance from community leaders, obtained oral consent for the baseline survey. Target respondents were randomly selected using the Kish method. Non-responses were recorded if the selected individual refused or was unavailable, except in cases of neuropsychiatric conditions, where a knowledgeable family member was allowed to serve as an informant. Written informed consent—including permission for audio recording—was obtained before each interview. Completed SCID-5-RV records were submitted electronically and promptly reviewed by a team leader, who provided immediate feedback for correction when necessary.

Quality control procedures included field supervision, logical data checks, audio-based review of interviews, and telephone follow-ups. All paper-based questionnaires underwent independent double data entry. Data verification involved identifying and correcting outliers, skip errors, and inconsistencies between reported symptoms and final diagnoses. Logical checks ensured the validity of key variables and the coherence between diagnostic criteria and assigned diagnoses. Notably, this study implemented a comprehensive, program-based verification system for all psychiatric diagnostic data for the first time, thereby enhancing consistency and traceability compared to manual review methods. For audio verification, the first two interviews conducted by each interviewer were reviewed, as well as 50% of interviews that resulted in a diagnosis and 16% of those without a diagnosis. Cases showing discrepancies between the audio recordings and the SCID-5-RV forms were discussed with the interviewer, and re-interviews were arranged when necessary. In addition, 20% of each interviewer's cases were randomly selected for telephone follow-up to confirm respondent identity. Following these rigorous quality assurance procedures, more than 95% of the interviews were verified to have met protocol standards, with only negligible residual errors.

Ethical clearance for this research was obtained from the Ethics Committee of Beijing Anding Hospital, Capital Medical University (Approval No.: MR-11-23-010737). All procedures followed the ethical principles outlined in the Declaration of Helsinki (1975, amended 2000) and

institutional regulations. Written informed consent was obtained from all participants before data collection, and they were notified of their right to withdraw from the study at any time without negative consequences.

Measurements

Survey tools

The diagnostic instrument used in this study was the adapted Chinese version of the SCID-5-RV, a semi-structured interview with established reliability and validity. This survey represents the first large-scale epidemiological study in China to employ the DSM-5 criteria, thereby enabling direct international comparisons. Both current and lifetime prevalence of mental disorders were assessed. However, the present analysis focuses solely on current diagnoses. This approach was adopted because insomnia disorder—a newly added diagnostic category in SCID-5-RV not present in the previous version (the Structured Clinical Interview for DSM-IV Axis I disorders, SCID [22–24])—is only evaluated as a current condition within the instrument. Consequently, the analyzed diagnostic categories included insomnia disorder, depressive disorders, bipolar and related disorders, anxiety disorders, obsessive–compulsive and related disorders, and alcohol use disorder.

A sociodemographic questionnaire was developed to collect information on monthly income, smoking habits, and chronic diseases. Chronic disease status was determined by self-report. In this study, respondents were classified as urban or rural based on the designation of their residential community, as defined by the National Bureau of Statistics. Specifically, urban areas refer to towns and cities characterized by a high density of non-agricultural populations and well-developed infrastructure, whereas rural areas comprise villages primarily engaged in agricultural activities. This approach aligns with established methodologies used in prior high-quality epidemiological studies in China [24].

The diagnostic process of DSM-5 insomnia disorder in SCID-5-RV

The diagnosis of insomnia disorder followed a sequential eight-step assessment:

- 1) Initial Screening: All respondents were asked, “Over the past 3 months, has a major concern of yours been that you are not getting enough good sleep or not feeling rested?” Those answering ‘yes’ were classified as having a ‘subjective sleep problem’. The “major concern” wording reflects the original SCID-5-RV interview item and was used to assess clinically significant subjective sleep dissatisfaction, which is consistent with DSM-5 Criterion A for insomnia

disorder, defined as a predominant complaint of dissatisfaction with sleep quantity or quality.

- 2) Symptom Evaluation: This required that respondents with a subjective sleep problem exhibit at least one of the following: difficulty initiating sleep (≥ 30 min), difficulty maintaining sleep, or early morning awakening (defined in accordance with DSM-5 as awakening at least 30 min before the scheduled time and before total sleep time reaches 6.5 h).
- 3) Functional Impairment Assessment: Respondents who fulfilled the symptom criteria were assessed to determine if their sleep disturbance caused clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- 4) Frequency Assessment: A minimum frequency of three nights per week for three months was required.
- 5) Opportunity for Sleep Assessment: It was confirmed that symptoms occurred despite adequate opportunity for sleep.
- 6) Primary Insomnia Assessment: Cases were evaluated to ensure symptoms were not exclusively attributable to the physiological effects of a substance.
- 7) Not attributable to coexisting conditions: In accordance with DSM-5 Criterion H, insomnia disorder was diagnosed when insomnia symptoms were judged by trained psychiatrists, not to be fully attributable to another mental disorder or medical condition, rather than excluding individuals with comorbid conditions.
- 8) Exclusion of Other Sleep Disorders: The disturbance was not better explained by another sleep–wake disorder.

Definition of insomnia subtype

Based on DSM-5 criteria, insomnia subtypes were defined as follows: initial insomnia (difficulty initiating sleep), middle insomnia (frequent or prolonged awakenings), and late insomnia (early-morning awakening with an inability to return to sleep). Participants reporting insomnia symptoms were correspondingly classified into these three subtypes according to their reported symptoms. The presentation of insomnia symptoms ranged from a single symptom to multiple co-occurring symptoms.

Statistical analyses

Prevalence estimates were calculated both with and without weights using the Statistical Analysis System (SAS, version 9.4). The unweighted prevalence was defined as the ratio of diagnosed cases to all participants assessed. The weighted prevalence, which accounted for sampling design, non-response, and post-stratification, was calculated with weights derived as follows: non-response

weights were the reciprocal of the estimated response probability from a logistic model, and post-stratification weights were calibrated to match the Beijing census data for distributions of gender, age, and urban–rural residence. The 95% confidence intervals (CIs) were calculated. Weighted Rao-Scott chi-square tests were used for descriptive assessment of survey-weighted associations, and no causal or directional inferences were intended. As a sensitivity analysis, we applied an alternative case definition of insomnia disorder that retained all DSM-5 diagnostic criteria except Criterion H (i.e., insomnia disorder defined without applying Criterion H). Statistical significance was declared for two-sided p -values < 0.05 .

Results

A total of 10,778 participants who completed the diagnostic interview constituted the final analytic sample, representing 87.1% of the initially interviewed cohort ($n = 12,379$) from 15,061 eligible households. Briefly, the sample was 50.9% male with a mean age of 46.0 ± 16.0 years. The majority (77.7%) were never or former smokers, and approximately one-third (32.9%) had a current chronic disease.

From insomnia symptoms to disorder: unraveling the diagnostic pathway

During the three months prior to the investigation, 1,961 respondents (17.7%) reported a subjective sleep problem (Fig. 1). Of these, 68% ($n = 1,333$) met the standardized criteria for DSM-5 insomnia disorder, constituting 10.7% of the total sample. The sequential application of criteria for functional impairment and clinical distress halved the number of eligible cases to 626. Subsequently, only 77% of these ($n = 484$) fulfilled the disease duration requirement. After excluding cases attributable to insufficient sleep opportunities or substance-related effects, 93% ($n = 450$) of the preceding cohort retained eligibility. Finally, 68 cases (15.1% of the 450) were excluded as their insomnia was fully explained by a pre-existing condition (41 by both mental disorders and chronic physical diseases, 20 only by mental disorders, and 7 only by chronic physical diseases), and one additional case was removed due to a better-explaining sleep disorder.

Prevalence of insomnia symptoms and disorder by sociodemographics characteristics and health

The weighted prevalence of clinically assessed insomnia symptoms was 10.7% (95% CI: 10.2–11.3), whereas 3.0% (95% CI: 2.7–3.3) met the criteria for DSM-5-diagnosed insomnia disorder. Both conditions shared significant associations with older age, being separated or widowed or divorced, unemployment or retirement, and chronic diseases (Table 1). Notably, female sex and lower educational attainment were associated with insomnia

symptoms but not with the full insomnia disorder. Neither condition was associated with place of residence, income, or smoking. The overall prevalence patterns and sociodemographic distributions of insomnia disorder were materially unchanged in this sensitivity analysis, suggesting that diagnostic attribution under Criterion H did not meaningfully bias the main findings (Table S3).

The subtypes of insomnia symptoms and disorder

Initial insomnia symptom demonstrated the highest weighted prevalence (8.1%) among the three symptom subtypes, followed by middle and late insomnia symptom (Table 2). At the symptom level, a single insomnia symptom was the most commonly reported profile (5.4%), ahead of any two symptoms (2.8%) and all three symptoms (2.5%). Among participants with only one symptom, initial insomnia was the most frequent. Of those endorsing exactly two symptoms, the combination of initial and middle insomnia was predominant.

A similar pattern was observed for insomnia disorder, with initial insomnia disorder again showing the highest prevalence (2.6%). Middle insomnia and early awakening disorder exhibited identical prevalence rates. In contrast to the symptom profiles, the most common disorder profile was the presence of all three symptoms (1.1%), followed by any two symptoms (1.0%) and a single symptom (0.9%).

Prevalence of insomnia disorder subtypes by sociodemographics characteristics and health

A distinct sociodemographic patterning was observed in the prevalence of insomnia disorder subtypes (Table 3). Females had a significantly higher prevalence of middle and late insomnia disorder than males, whereas no sex difference was found for initial insomnia disorder. An age-specific gradient was also evident: while all subtypes were most prevalent in the ≥ 60 year groups and least common in the 35–49 year group, the relative position of the 18–34 year group varied, ranking third for initial and late insomnia disorder but fourth for middle insomnia disorder. Furthermore, initial insomnia disorder was the dominant subtype in all age groups except the 50–59 year group, where middle insomnia disorder was most common.

Marital status was also significantly associated with prevalence patterns; middle and late insomnia disorder were highest in the separated/widowed/divorced group and lowest in the unmarried group, whereas initial insomnia disorder showed no significant variation. Education level was negatively associated with the prevalence of middle insomnia disorder but unrelated to initial or late insomnia disorder. Employment status further differentiated risk, with unemployed, retired, and homemaker individuals exhibiting significantly higher prevalence of

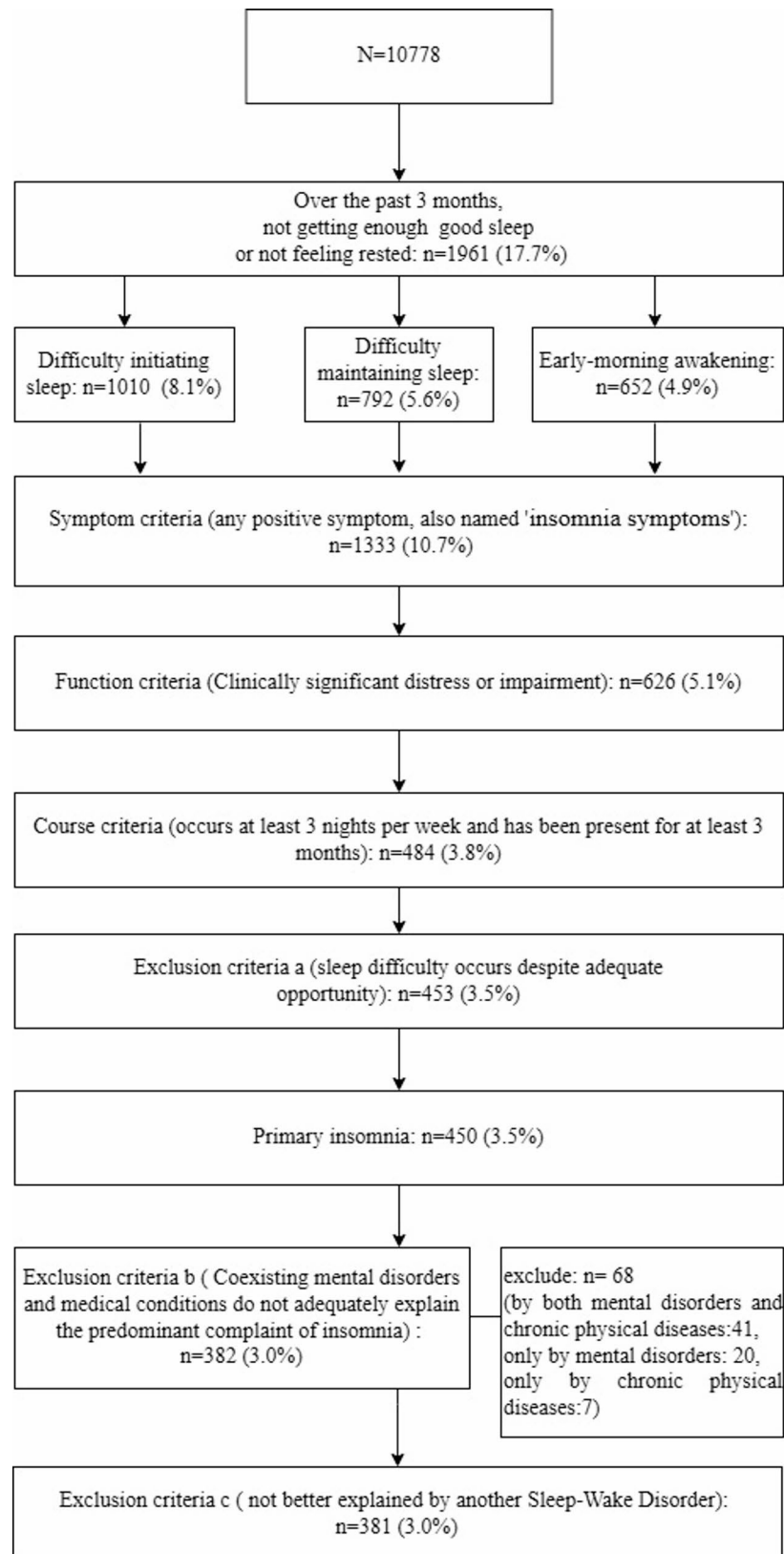
**Fig. 1** DSM-5 insomnia disorder diagnosis flow chart among adults in Beijing

Table 1 Sample description, prevalence of insomnia symptoms and disorder by sociodemographics characteristics and health

Variables	Total sample (%)	Insomnia symptoms [†]				Insomnia disorder [#]			
		n	weighted prevalence (% 95%CI)	χ^2	P	n	weighted prevalence (% 95%CI)	χ^2	P
Total sample	10,778(100)	1333	10.7(10.2–11.3)			381	3.0(2.7–3.3)		
Gender				6.571	0.010			0.618	0.432
Male	4326(50.9)	437	9.8(9.1–10.6)			123	2.8(2.4–3.3)		
Female	6452(49.1)	856	11.7(10.9–12.6)			258	3.2(2.8–3.7)		
Age (years)				52.967	< 0.0001			30.541	< 0.0001
18–34	1294(30.5)	100	8.4(7.5–9.4)			34	3.1(2.6–3.8)		
35–49	2725(29.7)	231	8.6(7.6–9.6)			46	1.9(1.5–2.5)		
50–59	2520(17.3)	297	11.1(9.8–12.6)			82	2.4(1.8–3.2)		
60–69	2744(13.4)	413	15.6(13.8–17.5)			119	3.9(3.0–5.0)		
≥ 70	1495(9.2)	292	17.9(15.6–20.4)			100	6.3(4.9–8.0)		
Place of residence				0.358	0.550			0.964	0.326
Urban	8777(87.5)	1076	10.9(10.3–11.5)			315	3.1(2.8–3.5)		
Rural	2001(12.5)	257	9.9(8.4–11.6)			66	2.5(1.8–3.5)		
Marital status				23.246	< 0.0001			6.805	0.033
Never married	626(16.1)	68	9.5(8.2–10.9)			21	2.9(2.2–3.8)		
Married or cohabitating	8840(16.1)	1010	10.2(9.5–10.8)			285	2.8(2.4–3.1)		
Separated, widowed, or divorced	1312(8.7)	255	18.3(15.9–20.9)			75	5.5(4.2–7.1)		
Education level				19.170	< 0.0001			3.397	0.183
Junior high school and below	4177(25.2)	620	14.0(12.8–15.4)			186	3.9(3.2–4.7)		
High school	2778(21.7)	331	11.1(9.8–12.4)			87	2.8(2.2–3.5)		
College or university or higher	3823 (53.1)	382	9.1(8.3–9.8)			108	2.7(2.3–3.2)		
Working status				28.347	< 0.0001			7.885	0.005
Working or student	4981(64.6)	462	8.8(8.2–9.5)			119	2.5(2.1–2.9)		
Unemployed, Retired or Homemaker	5797(35.4)	871	14.3(13.2–15.5)			262	4.0(3.5–4.7)		
Income [*]				0.680	0.409			0.001	0.984
≥ 5000 yuan/month	4373(55.2)	499	10.4(9.6–11.2)			148	3.0(2.6–3.5)		
< 5000 yuan/month	6370(44.8)	830	11.2(10.3–12.1)			233	3.0(2.6–3.6)		
Smoking [†]				0.361	0.548			0.394	0.530
No or ex-smoker	8811(77.7)	1115	10.6(9.9–11.3)			316	2.9(2.6–3.3)		
Current smoker	1934(22.3)	214	11.3(10.1–12.6)			65	3.3(2.7–4.1)		
Chronic diseases				97.073	< 0.0001			19.549	< 0.0001
No	6115(67.1)	453	7.3(6.7–7.9)			104	1.9(1.6–2.3)		
Yes	4663(32.9)	880	17.8(16.6–19.1)			277	5.2(4.6–6.0)		

Numbers were unweighted, but percentages and prevalence were weighted

Means multiple Rao-Scott χ^2 tests were used with the health controls as the reference

[†]Insomnia symptoms were clinically identified through psychiatrist-administered structured diagnostic interviews as the initial step of the DSM-5 diagnostic pathway

[#]Insomnia disorder was diagnosed when individuals fulfilled the full DSM-5 diagnostic criteria

^{*}Data missing from 35 respondents

all subtypes relative to working individuals or students. In contrast, no significant association was found for place of residence, income or smoking status with any insomnia subtype. Finally, the presence of chronic diseases was a consistent risk factor, with affected respondents showing a higher prevalence of all three subtypes compared to those without such conditions.

Co-occurrence of insomnia disorder and other mental disorders

To characterize the co-occurrence of insomnia disorder and other mental disorders in this cross-sectional sample, conditional prevalences were examined. Insomnia disorder was present in 14.4% of individuals with other mental disorders (most prevalent in depressive [21.4%], anxiety [19.2%], and alcohol-related disorders [10.9%]), while 31.3% of those with insomnia disorder had a comorbid mental disorder (primarily alcohol-related [13.4%], depressive [11.6%], and anxiety disorders [10.6%])

Table 2 The subtypes of insomnia symptoms and disorder

Subtypes of insomnia	Insomnia symptoms [†]		Insomnia disorder [#]	
	n	weighted prevalence (%;95%CI)	n	weighted prevalence (%;95%CI)
Any insomnia symptom	1333	10.7(10.2–11.3)	381	3.0(2.7–3.3)
Initial insomnia	1010	8.1(7.6–8.6)	320	2.6(2.3–2.9)
Middle insomnia	792	5.6(5.2–6.0)	266	1.8(1.6–2.1)
Late insomnia	652	4.9(4.5–5.3)	237	1.8(1.6–2.1)
Only one of the symptoms	574	5.4(5.0–5.9)	95	0.9(0.8–1.1)
Initial insomnia	372	3.7(3.3–4.0)	68	0.8(0.7–1.0)
Middle insomnia	134	1.0(0.9–1.3)	22	0.1(0.1–0.2)
Late insomnia	68	0.7(0.6–0.9)	6	0.02(0.01–0.07)
Only two of the symptoms	397	2.8(2.5–3.2)	128	1.0(0.9–1.2)
Initial + middle	175	1.1(1.0–1.4)	54	0.3(0.2–0.5)
Middle + late	121	0.9(0.7–1.1)	33	0.3(0.2–0.4)
Initial + late	101	0.8(0.7–1.0)	41	0.4(0.3–0.6)
Initial + middle + late	362	2.5(2.2–2.8)	157	1.1(0.9–1.3)

[†]Insomnia symptoms were clinically identified through psychiatrist-administered structured diagnostic interviews as the initial step of the DSM-5 diagnostic pathway

[#]Insomnia disorder was diagnosed when individuals fulfilled the full DSM-5 diagnostic criteria

(Table 4). The results indicated that the main patterns of co-occurrence estimates were robust to stratification by the sensitivity analysis (Table S4).

Discussion

As the first large-scale epidemiological study in China to employ a psychiatrist-administered, DSM-5-based diagnostic interview for insomnia, this research provides data of high validity. The rigorous methodology, still rare globally, identified that 17.7% of respondents reported subjective sleep problems, 10.7% reported insomnia symptoms, and 3.0% met the full DSM-5 criteria for insomnia disorder. Given that variations in assessment tools, sample composition, and diagnostic criteria significantly affect reported rates, direct comparisons with other studies are difficult. Consequently, any comparison of insomnia prevalence must be preceded by a critical examination of both historical changes in diagnostic standards and differences among concurrent diagnostic frameworks.

The evolution of insomnia diagnostic criteria from DSM-IV to DSM-5 and its consequences for epidemiological research

The transition from DSM-IV's "primary insomnia" to DSM-5's "insomnia disorder" introduced significant refinements across symptom, duration, and exclusion criteria, collectively influencing the estimation and comparability of prevalence rates.

In the domain of symptom evaluation, DSM-5 replaced ambiguous descriptions with specific, quantifiable metrics to enhance operational clarity and diagnostic consistency. For instance, the core symptoms of sleep onset and maintenance difficulties now require a quantitative threshold of more than 30 min. Additionally, the non-specific complaint of "nonrestorative sleep" was omitted, while "early-morning awakening" was introduced with a corresponding quantitative definition: awakening at least 30 min before the desired time and before achieving 6.5 h of total sleep. This methodological shift implies that the use of quantified criteria may lead to lower prevalence estimates compared to those derived from the less precise DSM-IV standards [13, 25, 26].

Regarding disease course, the required duration shifted from at least one month in DSM-IV to a stricter standard of ≥ 3 nights per week over ≥ 3 months in DSM-5.

Exclusion criteria were also substantially revised. DSM-5 [14] now explicitly excludes cases with insufficient sleep opportunity and no longer requires that insomnia occur "exclusively during the course of" another disorder, instead stipulating that comorbid conditions should not "adequately explain the predominant complaint." This shift facilitates the diagnosis of comorbid insomnia but introduces a methodological constraint: such exclusions depend on clinical judgment and cannot be applied in studies relying solely on self-report scales. Consequently, scale-based prevalence studies inherently omit these clinical nuances, complicating cross-manual comparisons.

Given these methodological discrepancies, we recommend that future epidemiological syntheses explicitly address these differences to ensure valid comparisons between estimates derived from DSM-IV and DSM-5 criteria.

Major classification systems for insomnia diagnosis

The diagnostic criteria for insomnia disorder are primarily defined by three classification systems: the DSM-5, the International Classification of Diseases, Eleventh Revision (ICD-11) [27], and the International Classification of Sleep Disorders, Third Edition, Text Revision (ICSD-3-TR) [28].

Insomnia is referred to as "insomnia disorder" in DSM-5, while both ICD-11 and ICSD-3-TR classify it under "chronic insomnia" and "chronic insomnia disorder," respectively. While the diagnostic criteria across these systems are largely consistent, several nuanced differences exist. Specifically, ICD-11 describes the frequency of insomnia symptoms only as "several times" per week, in contrast to the more explicit requirement of "at least three times" stipulated in DSM-5-TR. In addition, ICSD-3-TR includes two supplementary symptoms related to sleep initiation: reluctance to go to bed at an appropriate

Table 3 Prevalence of insomnia disorder subtypes by sociodemographics characteristics and health

	Initial insomnia disorder				Middle insomnia disorder				Late insomnia disorder			
	n	Weighted prevalence (% 95%CI)	χ ²	P	n	weighted prevalence (% 95%CI)	χ ²	P	n	weighted prevalence (% 95%CI)	χ ²	P
Gender			0.735	0.391			3.999	0.046			4.941	0.026
Male	102	2.4(2.1–2.9)			76	1.4(1.1–1.7)			70	1.3(1.1–1.7)		
Female	218	2.8(2.4–3.3)			190	2.2(1.9–2.7)			167	2.3(1.9–2.7)		
Age (years)			35.168	<0.0001			34.529	<0.0001			31.019	<0.0001
18–34	33	3.1(2.6–3.7)			12	1.1(0.8–1.6)			15	1.4(1.0–1.9)		
35–49	32	1.4(1.0–1.8)			25	1.1(0.8–1.5)			25	1.2(0.9–1.7)		
50–59	63	1.8(1.3–2.5)			64	2.0(1.4–2.7)			51	1.4(0.9–2.0)		
60–69	101	3.3(2.5–4.3)			88	2.6(1.9–3.5)			75	2.4(1.7–3.3)		
≥ 70	91	5.6(4.4–7.2)			77	4.8(3.6–6.3)			71	4.8(3.6–6.3)		
Place of residence			1.688	0.194			0.019	0.891			0.260	0.610
Urban	265	2.7(2.4–3.0)			207	1.8(1.5–2.1)			192	1.9(1.6–2.1)		
Rural	55	2.1(1.4–3.0)			59	1.9(1.3–2.7)			45	1.5(1.0–2.3)		
Marital status			4.349	0.114			14.066	0.001			16.733	0.0002
Never married	20	2.9(2.2–3.8)			7	0.5(0.3–1.0)			5	0.6(0.3–1.1)		
Married or cohabitating	237	2.4(2.0–2.7)			202	1.8(1.5–2.1)			174	1.7(1.5–2.1)		
Separated, widowed, or divorced	63	4.3(3.2–5.8)			57	3.9(2.8–5.3)			58	4.3(3.2–5.8)		
Education level			3.134	0.0209			7.445	0.024			3.510	0.173
Junior high school and below	161	3.3(2.7–4.1)			149	2.7(2.2–3.4)			127	2.4(1.9–3.0)		
High school	75	2.5(2.0–3.3)			59	1.6(1.1–2.1)			52	1.7(1.2–2.3)		
College or university or higher	84	2.3(1.9–2.7)			58	1.4(1.2–1.8)			58	1.6(1.3–1.9)		
Working status			7.977	0.005			10.850	0.001			10.856	0.001
Working or student	94	2.1(1.8–2.4)			71	1.3(1.0–1.5)			67	1.3(1.0–1.6)		
Unemployed, Retired or Homemaker	226	3.6(3.0–4.2)			195	2.8(2.3–3.3)			170	2.7(2.3–3.3)		
Income*			0.067	0.795			0.637	0.425			0.357	0.550
≥ 5000 yuan/month	122	2.6(2.2–3.2)			90	1.7(1.4–2.0)			88	1.7(1.4–2.1)		
< 5000 yuan/month	198	2.7(2.2–3.0)			176	2.0(1.6–2.4)			149	1.9(1.6–2.3)		
Smoking*			1.581	0.209			2.700	0.100			2.805	0.094
No or ex-smoker	261	2.5(2.1–2.8)			231	2.0(1.7–2.3)			202	1.9(1.7–2.2)		
Current smoker	59	3.2(2.6–4.0)			35	1.2(0.9–1.8)			35	1.3(0.9–1.9)		
Chronic diseases			21.162	<0.0001			16.602	<0.0001			16.750	<0.0001
No	86	1.7(1.4–2.0)			62	1.0(0.8–1.3)			54	1.0(0.8–1.3)		
Yes	234	4.5(3.9–5.3)			204	3.3(2.8–4.0)			183	3.3(2.8–4.0)		

Numbers were unweighted, but percentages and prevalence were weighted

Means multiple Rao-Scott χ^2 tests were used with the health controls as the reference

*Data missing from 35 respondents

Table 4 Co-occurrence of between insomnia disorder and other mental disorders

Mental disorders	Co-occurring cases [†]	Other mental disorder co-occurrence insomnia disorder [#]	Insomnia disorder co-occurrence other mental disorder [*]
	Frequency, n	Weighted prevalence (%;95% CI)	Weighted prevalence (%;95% CI)
Depressive Disorders	42	21.4(15.4–27.5)	11.6(8.1–15.0)
Anxiety Disorders	27	19.2(13.4–25.0)	10.6(7.2–13.9)
Alcohol-related Disorders	24	10.9(7.9–14.0)	13.4(9.7–17.1)
Obsessive–Compulsive and Related Disorders	5	0.5(0.01–7.3)	0.5(0.01–1.3)
Bipolar and Related Disorders	0	NA	NA
Any of the five studied mental disorders	79	14.4(12.0–17.2)	31.3(26.5–36.3)

NA not applicable

[†]The “Co-occurring cases” column indicated the number of respondents who currently met diagnostic criteria for both insomnia disorder and at least one additional mental disorder[#]The “Other mental disorder co-occurrence insomnia disorder” column represented the weighted prevalence of individuals with the specified mental disorder who also had insomnia disorder^{*}The “Insomnia disorder co-occurrence other mental disorder” column represented the weighted prevalence of individuals with insomnia disorder who also had the specified mental disorder

time, and difficulty falling asleep without intervention from a parent or caregiver—a presentation more commonly observed in children or older adults with severe impairment. Furthermore, ICSD-3-TR elaborates on the clinical significance criterion by listing nine specific manifestations of distress or functional impairment, thereby enhancing operational clarity and practical applicability.

Therefore, epidemiological findings on insomnia disorder are broadly comparable across these three diagnostic systems. However, the relatively high prevalence rates [29] associated with ICSD-3 [30] can be attributed to its less stringent exclusion criteria for medical, mental, and substance-related comorbidities compared to ICSD-3-TR and DSM-5. Additionally, owing to nearly identical diagnostic criteria of insomnia disorder in the ICD-10 Classification of Mental and Behavioural Disorders [31] and DSM-IV, the prevalence diagnosed by ICD-10 is also higher [26] than that determined by the stricter DSM-5 criteria.

Comparisons of insomnia prevalence and subtypes

The estimated prevalence of insomnia in China shows considerable variation across regions and assessment methodologies, making it difficult to determine whether this reflects true regional disparities or methodological

heterogeneity. For instance, a nationwide online survey employing convenience sampling and the Insomnia Severity Index (ISI) during February–March 2020 reported that 29.2% of Chinese adults experienced insomnia symptoms [32]. A subsequent large-sample, cross-sectional study in Guangdong Province (October–December 2022), which also used the ISI, revealed a similar prevalence of 25.2% [10]. In contrast, a smaller-scale study conducted in Macau, China (July–September 2022) using the same scale found a markedly higher prevalence of 49.0% [11]. Meanwhile, a November 2023 meta-analysis of the general Chinese population, which utilized the Chinese version of the Pittsburgh Sleep Quality Index (PSQI), reported a pooled prevalence of 19.0% [33].

Internationally, findings consistent with this study have been reported. For instance, the 30-day prevalence of individuals meeting all DSM-5 criteria for insomnia disorder was 3.0% in the general population of Qatar [20]. However, several studies have reported a higher prevalence of insomnia disorder than that found in the present study. For example, the one-year prevalence of subjective insomnia among adults in five European countries in 2020 ranged between 21.0% and 26.8%, while the prevalence of insomnia disorder—defined according to stricter diagnostic criteria—was 6.0% [34]. This discrepancy may be attributed to methodological variations, such as the application of a shorter minimum symptom duration (e.g., 1.5 months) and reliance on self-report instruments such as the ISI. Meanwhile, a cross-sectional survey of a Norwegian general population (≥ 40 years) using the revised Bergen Insomnia Scale (BIS) revealed a DSM-5 insomnia prevalence of 8.5% [26]. Similarly, the prevalence of insomnia disorder identified solely using the Sleep Condition Indicator (SCI) [35]—a DSM-5-based screening tool with a cutoff score of 16—was also relatively higher [36]. A meta-analysis further highlighted that the pooled prevalence from studies using diagnostic interviews to apply DSM criteria was 12.4% [18], a rate likely elevated due to the inclusion of studies using self-assessment tools. Such instruments generally lack exclusion criteria and may capture individuals who do not fully meet all diagnostic requirements regarding symptom severity and duration.

Regarding insomnia subtypes, this study confirmed that difficulty falling asleep was the most prevalent single symptom [26, 37, 38], while the combination of sleep initiation, maintenance, and early morning awakening difficulties constituted the most frequent symptom profile in DSM-5 insomnia disorder [14, 37].

Correlates of insomnia

This study found that both insomnia symptoms and disorder were significantly associated with older age, non-married status, unemployment or retirement, and

chronic diseases. Furthermore, female sex and lower educational attainment were specific risk factors for insomnia symptoms. These findings align with existing literature [3, 10, 38–40].

Although urban residents exhibited a higher prevalence of insomnia than their rural counterparts, the difference was not statistically significant. This may be attributed to the high urbanization rate in Beijing (approximately 88% in 2021), which has led to a convergence of lifestyle factors—such as work stress, electronic device usage, and environmental exposures—between urban and rural areas [41, 42]. Consequently, sleep-related risk profiles have become increasingly homogeneous across these populations.

Similarly, a non-significant trend toward higher insomnia prevalence was observed among smokers compared to non-smokers. This may be due to the modest effect size of smoking on insomnia ($OR \approx 1.07$, as reported in meta-analyses) [43], which requires large samples or refined measures to detect. The lack of stratification by smoking intensity or timing—particularly night-time smoking, which is more strongly linked to sleep disturbances—may have further limited our ability to identify a significant association [44].

Finally, monthly income exceeding 5,000 yuan was not associated with insomnia severity or disorder prevalence. This null finding may reflect substantial within-category heterogeneity [45], the limited relevance of personal income relative to broader socioeconomic determinants (e.g., housing quality) [46], or potential reporting bias due to the sensitivity of financial data. Future studies should employ multidimensional socioeconomic metrics—such as perceived financial strain and disposable income—and consider subgroup analyses to better elucidate these complex relationships.

Co-occurrence: insomnia and mental disorders

This study reinforces the evidence for co-occurrence of insomnia disorder and other mental disorders, which arises from shared neurobiological and psychological pathways [47]. Furthermore, the study found that alcohol-related disorders were the most prevalent comorbid mental condition among individuals with insomnia disorder. One potential explanation for this is that affected individuals may initiate alcohol consumption as a maladaptive coping strategy to induce sleep, which may, over time, progress to a substance use disorder [48]. In addition, a high prevalence of comorbid insomnia disorder was observed in individuals with depressive or anxiety disorders. This finding is supported by evidence from other epidemiological studies [10, 39, 49] and real-world clinical data [50, 51]. Taken together, these results mandate the active treatment of insomnia disorder, as it has

substantial relevance for clinical outcomes and broader public health priorities [5, 50].

Limitations

The interpretation of our findings should consider several limitations. The cross-sectional design precludes causal inference and may be susceptible to recall bias. Furthermore, pragmatic constraints necessitated the exclusion of certain diagnostic categories (e.g., specific Substance-Related and Schizophrenia-Spectrum Disorders), potentially limiting the comprehensiveness of the documented psychiatric comorbidity. In addition, diagnostic attribution of insomnia disorder under DSM-5 Criterion H involves clinical judgment and may be subject to some degree of misclassification. Nevertheless, sensitivity analyses without applying Criterion H showed consistent results, indicating that the core findings are robust to potential attribution-related bias. Finally, the absence of a full range of known risk factors restricted the etiological exploration. Despite these limitations, this study provides robust, novel data as the first large-scale epidemiological investigation in China to employ the rigorous, yet globally uncommon, methodology of psychiatrist-administered, DSM-5-based diagnostic interviews for insomnia. Accordingly, longitudinal research and more detailed multivariable analyses are warranted to extend beyond the descriptive associations identified in this cross-sectional study.

Conclusion

Our findings reveal a pronounced “iceberg” of insomnia, wherein a 10% prevalence of symptoms dwarfs the 3% with a formal diagnosis, which reveals a vast subclinical and high-risk cohort. Given the significant comorbidity with mental disorders, this disparity positions insomnia management as a crucial public health lever. We therefore advocate for a strategic shift toward population-level screening and early intervention, supported by the adoption of standardized diagnostics for reliable data. In clinical practice, this means implementing routine sleep assessments in psychiatric care and mental health evaluations for insomnia patients, enabling integrated, transdiagnostic models as a pivotal preventive measure.

Abbreviations

DSM-5	Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition
DSM-IV	Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition
DSM-5-TR	Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision
BMHS	Beijing Mental Health Survey
SCID-5-RV	Structured Clinical Interview for DSM-5 Disorders, Research Version
PPS	Probability Proportionate to Size
CAPI	Computer-assisted personal interviewing
PAPI	Pen-and-paper personal interviews

SCID	Structured Clinical Interview for DSM-IV Axis I disorders
CI	Confidence interval
ICD-11	International Classification of Diseases, Eleventh Revision
ICSD-3-TR	International Classification of Sleep Disorders, Third Edition, Text Revision
ISI	Insomnia Severity Index
PSQI	Pittsburgh Sleep Quality Index
BIS	Bergen Insomnia Scale
SCI	Sleep Condition Indicator
SAS	Statistical Analysis System

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12889-026-26453-x>.

Supplementary Material 1.

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Authors' contributions

LS, PL and LZ designed the study; LS, YZ, AC, HD and LZ involved in data collection and supervision; LS conducted statistical analysis; LS, XZ and LZ contributed to the writing of the manuscript and provided English proofreading. All authors reviewed the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available but are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted following approval from the Ethics Committee of Beijing Anding Hospital, Capital Medical University (Approval No: MR-11-23-010737) and in compliance with the Declaration of Helsinki and institutional rules. Voluntary informed consent was secured from all participants before any study procedures, and they were assured of their right to withdraw at any time without consequence.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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